

Jointly Non-Degenerate Products and Open Multiplication in Hölder–Lipschitz Algebras

Full solution packet for arXiv:2301.02320, Question 1

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Status. `candidate_full_solution_likely_valid`. In the compact metric setting, the Lipschitz-algebra part of Question 1 has a complete answer: only finite spaces give the requested approximation property. A stronger theorem also classifies open multiplication and covers every big Hölder order $0 < \alpha \leq 1$.

1 Source question

Kania and Maślany introduce a unital Banach $*$ -algebra property they call *approximability by jointly non-degenerate products*: for all f, g and $\varepsilon > 0$, there are jointly non-degenerate f', g' within ε such that $f'g' = fg$ [1]. Their Question 1 on printed page 21 asks:

“What about algebras of Lipschitz functions on zero-dimensional compact spaces?”

5. OPEN PROBLEMS

In the light of Theorem A let us pose the following question.

Question 1. What are further examples of (dual) Banach algebras that are approximable by jointly non-degenerate elements? What about algebras of Lipschitz functions on zero-dimensional compact spaces?

We answer this for the standard big Lipschitz algebra on a compact metric space, and simultaneously for its Hölder variants.

2 Definitions and proof intuition

Let (X, d) be compact metric and $0 < \alpha \leq 1$. Write

$$\text{Lip}_\alpha(X) = \left\{ f : X \rightarrow \mathbb{C} : [f]_\alpha := \sup_{x \neq y} \frac{|f(x) - f(y)|}{d(x, y)^\alpha} < \infty \right\}, \quad \|f\|_\alpha = \|f\|_\infty + [f]_\alpha.$$

This is a unital Banach $*$ -algebra under pointwise operations. A pair f, g is *jointly non-degenerate* when $|f|^2 + |g|^2$ is invertible. In particular, such a pair has no common zero.

Proof intuition. At any accumulation point o , choose points $x_n \rightarrow o$ on geometrically shrinking scales. Give two functions alternating radial Hölder slopes: one has value $d(x_n, o)^\alpha$ on even indices and zero on odd indices, and the other has the reverse pattern. McShane extension realizes this data on all of X . Any nearby exact factorization of their product by a jointly non-degenerate pair must select one nonzero factor at o . That factor stays nonzero on a tail, forcing the other factor to vanish on that tail; one parity then retains a Hölder error quotient exactly equal to one.

The same alternating slopes obstruct open multiplication. A nonzero constant perturbation of the product would force both factors to be nonzero at o . The even and odd scales instead produce two mutually contracting inequalities for those two base values, forcing both to vanish.

3 Complete classification

Theorem 1. *Let (X, d) be a compact metric space and let $0 < \alpha \leq 1$. The following are equivalent:*

1. X is finite;
2. $\text{Lip}_\alpha(X)$ is approximable by jointly non-degenerate products;
3. pointwise multiplication

$$m : \text{Lip}_\alpha(X) \times \text{Lip}_\alpha(X) \longrightarrow \text{Lip}_\alpha(X), \quad m(f, g) = fg,$$

is open.

In particular, every infinite zero-dimensional compact metric space gives a negative answer to the Lipschitz-algebra part of Question 1.

We first isolate the alternating construction.

Lemma 1. *If X is infinite, there are $o, x_1, x_2, \dots \in X$, positive numbers $r_n = d(x_n, o) \downarrow 0$, and real $F, G \in \text{Lip}_\alpha(X)$ such that $r_{n+1} \leq r_n/3$ and, writing $\rho_n = r_n^\alpha$,*

$$F(o) = G(o) = 0, \quad (F(x_n), G(x_n)) = \begin{cases} (\rho_n, 0), & n \text{ even}, \\ (0, \rho_n), & n \text{ odd}. \end{cases}$$

Proof. An infinite compact metric space has an accumulation point o and distinct points tending to it. Passing to a subsequence gives $r_{n+1} \leq r_n/3$. On the closed subset $S = \{o, x_1, x_2, \dots\}$, prescribe the displayed values. If $n < m$, then

$$d(x_n, x_m) \geq r_n - r_m \geq \frac{2}{3}r_n.$$

The change of either prescribed function between x_n and x_m is at most r_n^α , while its change between x_n and o is at most r_n^α . Thus each is Lipschitz on the metric space (S, d^α) , with constant at most $(3/2)^\alpha$. Since d^α is a metric, the real McShane extension theorem extends both functions to X without increasing that constant [2]. $\square$

Proof of Theorem 1: failure of (2) on infinite X . Take F, G from Lemma 1. Suppose that $f, g \in \text{Lip}_\alpha(X)$ satisfy

$$fg = FG, \quad \|f - F\|_\alpha < \frac{1}{2}, \quad \|g - G\|_\alpha < \frac{1}{2},$$

and are jointly non-degenerate. Since $F(o)G(o) = 0$, we have $f(o)g(o) = 0$; joint non-degeneracy says that exactly one of $f(o), g(o)$ is zero.

Suppose first that $f(o) \neq 0$, so $g(o) = 0$. Continuity gives $f(x_n) \neq 0$ for every sufficiently large n . Since $f(x_n)g(x_n) = F(x_n)G(x_n) = 0$, it follows that $g(x_n) = 0$ on this tail. For odd n in the tail,

$$\frac{|(g - G)(x_n) - (g - G)(o)|}{d(x_n, o)^\alpha} = \frac{\rho_n}{r_n^\alpha} = 1,$$

contradicting $[g - G]_\alpha < 1/2$. If $g(o) \neq 0$, the even subsequence gives the identical contradiction for $f - F$. Therefore (2) fails. $\square$

Proof of Theorem 1: failure of (3) on infinite X . Again take F, G from Lemma 1, and fix $0 < \varepsilon < 1/2$. We prove that no nonzero constant perturbation $FG + t$ can be factored as ab with

$$\|a - F\|_\alpha < \varepsilon, \quad \|b - G\|_\alpha < \varepsilon. \quad (1)$$

Set $a_0 = a(o)$ and $b_0 = b(o)$. If such a factorization existed, then $a_0b_0 = t$. Define

$$u_n = (a - F)(x_n) - (a - F)(o), \quad v_n = (b - G)(x_n) - (b - G)(o).$$

By (1), $|u_n|, |v_n| \leq \varepsilon\rho_n$.

For even n , $a(x_n) = a_0 + \rho_n + u_n$ and $b(x_n) = b_0 + v_n$. Since both $FG(x_n)$ and $FG(o)$ vanish, subtracting $a_0b_0 = t$ from $a(x_n)b(x_n) = t$ and dividing by ρ_n gives

$$0 = b_0 + a_0 \frac{v_n}{\rho_n} + b_0 \frac{u_n}{\rho_n} + (\rho_n + u_n) \frac{v_n}{\rho_n}.$$

Letting even $n \rightarrow \infty$ yields

$$(1 - \varepsilon)|b_0| \leq \varepsilon|a_0|. \quad (2)$$

The odd subsequence, with the roles reversed, yields

$$(1 - \varepsilon)|a_0| \leq \varepsilon|b_0|. \quad (3)$$

As $\varepsilon < 1/2$, (2)–(3) force $a_0 = b_0 = 0$, contradicting $a_0b_0 = t \neq 0$.

If multiplication were open at (F, G) , the image of the product of the two ε -balls would contain a ball about FG . But constants $t \neq 0$ have $\|t1\|_\alpha = |t|$ and can be arbitrarily small, contradicting the preceding paragraph. Thus multiplication is not open. $\square$

Proof of Theorem 1: finite X . Suppose X is finite. Then $\text{Lip}_\alpha(X) = \mathbb{C}^X$ algebraically and its Hölder norm is equivalent to the supremum norm. Given f, g and a desired accuracy, leave both unchanged except at coordinates where $f(x) = g(x) = 0$; there, replace $f(x)$ by a sufficiently small fixed nonzero scalar and keep $g(x) = 0$. The product is unchanged, the resulting pair has no common zero and hence is jointly non-degenerate, and finite-dimensional norm equivalence makes the perturbation arbitrarily small. This proves (2).

For (3), scalar multiplication $\mathbb{C}^2 \rightarrow \mathbb{C}$ is open: away from $(0, 0)$ this follows from a nonzero partial derivative, and at $(0, 0)$ it follows by taking square roots. Coordinatewise multiplication on $\mathbb{C}^X$ is a finite product of these open maps, hence is open. Equivalent finite-dimensional norms do not change openness. $\square$

4 Scope, verification, and novelty

The theorem includes the ordinary big Lipschitz algebra at $\alpha = 1$ and is unchanged if the standard max norm replaces the sum norm. The alternating functions have small-scale Hölder quotients equal to one, so they need not belong to the little-Lipschitz subalgebra; no claim is made for that distinct space. A Lipschitz algebra also requires a metric, so nonmetrizable compacta without specified compatible metric data are outside the statement.

The included script checks the alternating construction on 80-level explicit models for five Hölder orders and checks the contraction coefficient in the non-openness proof. It returned **PASS**; the proof is exact and does not depend on computation.

Bounded searches on 11 August 2026 used the exact Question 1 wording, the paper title and authors, and combinations of *jointly non-degenerate products*, *Lip(X)*, *Hölder algebra*, and *open multiplication*. They found the source, its published version, talks reproducing the question, and general Lipschitz-algebra background, but no later answer or this finite-space classification. This is bounded evidence, not exhaustive novelty certification.

Human-review recommendation. Check the subsequence/McShane step, the forced-zero tail argument, and the passage to inequalities (2)–(3). The main scope question is whether the source intends the standard big Lipschitz algebra; under that standard reading the classification is complete.

References

- [1] T. Kania and N. Mařłany, *Differential embeddings into algebras of topological stable rank 1*, arXiv:2301.02320; Proc. Roy. Soc. Edinburgh Sect. A, doi:10.1017/prm.2024.108.
- [2] E. J. McShane, *Extension of range of functions*, Bull. Amer. Math. Soc. 40 (1934), 837–842.